YASKAWA

# YRC1000 & YRC1000micro OPTIONS INSTRUCTIONS

**FOR: MS3D CORTEX CONFIGURATION**

Upon receipt of the product and prior to initial operation, read these instructions thoroughly, and retain for future reference.

Have the following information available when contacting the YASKAWA Representative:

- System
- Primary Application
- Software Version *(Located on Programming Pendant by selecting: {Main Menu} - {System Info} - {Version})*
- Warranty ID *(Located on Robot Controller)*
- Robot Serial Number *(Located on Manipulator data plate)*
- Robot Sales Order Number *(Located on Robot controller data plate)*

MANUAL NO.
194223-1CD ⓪

**The Americas YASKAWA Representative**

- 24-hour Telephone Number: **(937) 847-3200**
  Use for urgent or emergency needs for technical support, service and/or replacement parts
- Routine Technical Inquiries: *techsupport@motoman.com*
  Allow up to 36 hours for response

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# Safety

## For Your Safety

Robots generally have requirements which are different from other manufacturing equipment, such as larger working areas, high-speed operation, rapid arm movements, etc., which can pose safety hazards.

Read and understand the instruction manuals and related documents, and observe all precautions in order to avoid the risk of injury to personnel and damage to equipment.

Carelessness contributes to serious accidents in the work area.

It is the user's responsibility to ensure that all local, state, and national codes, regulations, rules, or laws relating to safety and safe operating conditions are met and followed.

**DANGER**

- Teaching, operations, and maintenance of the Robot must conform to:
  - Industrial Safety and Health Law
  - Order for Enforcement of the Industrial Safety and Health Law
  - Industrial Safety and Health Regulations
  - Technical Standards for Electrical Facilities

  Other related laws and regulations are:
  - Occupational Safety and Health Act in USA
  - Factory Act (Gewerbeordnung) in Germany
  - Health and Safety at Work, etc. Act in UK
  - EC Machinery Directive 2006/42/EC
- Prepare:
  - SAFETY WORK REGULATIONS
    based on concrete policies for safety management complying with related laws and regulations.
- Observe:
  - JIS B 8433-1: 2015 "Robots for industrial environments-Safety requirements" (ISO 10218-1: 2011) for safe operation of the robot. (JIS B 8433 is for Japan only)
- Reinforce:
  - SAFETY MANAGEMENT SYSTEM
    by designating authorized operators and safety managers for the Robot, as well as giving continuing safety education and training.
- Teaching, operation, and maintenance of the Robot are specified as "Hazardous Operations" in the Industrial Safety and Health Act (for Japan only).
  Personnel engaged in these operations must receive special training offered by YASKAWA.

**DANGER**

- Personnel engaged in operation, maintenance, or management of the Robot must receive required training before using the Robot.

For more information on training, contact a YASKAWA representative.

- Make sure to have and follow all manuals, read them thoroughly and understand the contents of them.

Confirm that you have all required manuals. If any of the manuals are missing, contact a YASKAWA representative.

- Read and understand these instructions throughly before installing, operating, or maintaining the Robot.

Any matter not described in this manual must be regarded as “prohibited” or “improper”.

We suggest that you obtain and review a copy of the ANSI/RIA National Safety Standard for Industrial Robots and Robot Systems (ANSI/RIA R15.06-2012). You can obtain this document from the Robotic Industries Association (RIA) at the following address:

Robotic Industries Association
900 Victors Way
P.O. Box 3724
Ann Arbor, Michigan 48106
TEL: (734) 994-6088
FAX: (734) 994-3338
www.roboticsonline.com

Ultimately, well-trained personnel are the best safeguard against accidents and damage that can result from improper operation of the equipment. The customer is responsible for providing adequately trained personnel to operate, program, and maintain the equipment.

We recommend approved YASKAWA training courses for all personnel involved with the operation, programming, or repair of the equipment.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

**WARNING**

- Safe operation of this equipment is the users responsibility.
  - The conditions under which the equipment will be operated safely should be reviewed by the user. The user must be aware of the various national codes, ANSI/RIA R15.06-2012 safety standards, and other local codes that may pertain to the installation and use of this equipment.

Not following all national codes, safety standards and local codes can result in death or serious injury

- Additional safety measures for personnel and equipment may be required depending on system installation, operation, and/or location.
  - The following safety equipment is provided as standard:
    - Safety barriers
    - Door interlocks
    - EMERGENCY STOP button

Not providing additional safety measures as required can result in death or serious injury.

- Check all safety equipment frequently for proper operation.
  - Repair or replace any non-functioning safety equipment immediately.

If safety equipment does not operate properly death or serious injury can result.

**CAUTION**

- Only trained personnel familiar with the operation, manuals, electrical design, and interconnections of this equipment should program, or maintain the system.

Any personnel involved with the operation of the equipment must understand potential dangers of operation.

**NOTICE**

- The drawings and photos in this manual are examples. Differences may exist between them and the delivered product.
- YASKAWA may modify this model without notice due to product improvements, modifications, or changes in specifications. If such modification is made, the manual number will also be revised.
- Some operations require standard passwords and while others require special passwords.
- If a manual is damaged or lost, contact a YASKAWA representative to order a new copy. Make sure to tell the YASKAWA representative the Manual Number listed on the front cover.

## Notes for Safe Operation

Read this manual carefully before installing, operating, maintaining, or inspecting the system.

In this instruction, Safe Operations are classified as “DANGER”, “WARNING”, “CAUTION” or “NOTICE”.

| | |
|---|---|
| DANGER | Indicates an imminently hazardous situation which, if not avoided, **WILL result in death or serious injury**. |
| WARNING | Indicates a potentially hazardous situation which, if not avoided, **MAY result in death or serious injury**. |
| CAUTION | Indicates a hazardous situation, which if not avoided, **MAY result in minor to moderate injury**. |
| CAUTION | Indicate a situation which if not avoided **may result in equipment damage.** |
| NOTICE | Indicates **practices not related to personal injury**. |

| NOTICE |
|---|
| To ensure safe and efficient operation at all times, be sure to follow all instructions, even if not designated as “DANGER”, “WARNING” or “CAUTION”. |

## Definition of Terms Used Often in This Manual

The Robot is the YASKAWA industrial robot product.

The Robot usually consists of a Robot, Controller, Programming Pendant, and Robot cables.

In this manual, the equipment is designated as follows:

| Equipment | Manual Designation |
|---|---|
| YRC1000 & YRC1000micro controller | Controller |
| YRC1000 & YRC1000micro Programming Pendant | Programming Pendant |
| MS3D Cortex Configuration Manipulator | Robot |
| Cable between the Robot and the Controller | Robot cable |
| Positioner | Positioner |

Descriptions of the Programming Pendant keys, buttons, and displays are shown as follows:

| Equipment | | Manual Designation |
|---|---|---|
| Programming Pendant | Character Keys /Symbol Keys | The keys which have characters or symbols printed on them are denoted with [ ].<br>e.g. [ENTER] |
| | Axis Keys /Numeric Keys | [Axis Key] and [Numeric Key] are generic names for the keys for axis operation and number input. |
| | Keys pressed simultaneously | When two keys are to be pressed simultaneously, the keys are shown with a “+” sign between them, e.g. [SHIFT]+[COORD]. |
| | Mode Switch | Mode Switch can select three kinds of modes that are denoted as follows: REMOTE, PLAY or TEACH.<br>(The switch names are denoted as symbols) |
| | Button | The three buttons on the upper side of the Programming Pendant are denoted as follows: START, HOLD, or EMERGENCY STOP.<br>(The button names are denoted as symbols) |
| | Displays | The menu displayed in the Programming Pendant is denoted with { }.<br>e.g. {JOB} |

## Registered Trademark

In this manual, names of companies, corporations, or products are trademarks, registered trademarks, or bland names for each company or corporation. The indications of ® and TM are omitted.

# YASKAWA Representative Information

The following information is for a YASKAWA representative in America, for other locations refer to the back cover.

**(937) 847-3200**

For **routine** technical inquiries, you can also contact a YASKAWA representative at the following e-mail address:

techsupport@motoman.com

When using e-mail to contact a YASKAWA representative, please provide a detailed description of your issue, along with complete contact information. Please allow approximately 24 to 36 hours for a response to your inquiry.

| NOTICE |
| --- |
| Please use e-mail for routine inquiries only. If you have an urgent or emergency need for service, replacement parts, or information, you must contact a YASKAWA representative at the telephone number shown above or on the back cover. |

Please have the following information ready before calling a YASKAWA representative:

- System — MS3D Cortex Configuration
- Manipulator _____
- Positioner _____
- Primary Application _____
- Robot Controller _____
  *YRC1000 & YRC1000micro*
- Software Version _____
  *Access this information on the Programming Pendant's LCD display screen by selecting {MAIN MENU} - {SYSTEM INFO} - {VERSION}*
- Robot Serial Number _____
  *Located on the Manipulator data plate*
- Robot Sales Order Number _____
  *Located on the Robot Controller data plate*
- Warranty ID: _____
  *Located on Robot Controller*

# 1 Introduction

This product is based on Cortex Recognition software written by Recognition Robotics, Inc. and provides the MS3D Cortex MotoPlus application for the Robot Controller to communicate with the Cortex PC based vision guidance software package.

- Does not require structured lighting.
- Features a very easy to use interface
  - Does not require complex programming or calibration.
  - Determines the camera position which an object was trained providing a common reference point between the robot and the camera.
- Utilizes a single mobile camera.
- **The Cortex Recognition user manual should be consulted for details of how to use the program. The manual is accessible from the Programs → Recognition Robotics, Inc. selection.**
- This application requires but does not include an “embedded” PC. The PC needs to include an Ethernet port and a PoE (Power over Ethernet) port to communicate with and power a Baumer camera.
- The camera typically mounts on the robot and a TCP and reference frame being set up relative to the camera.
- The first step is to train or teach the Cortex Recognition software to recognize the universe of parts. The software assigns each part a unique number. YASKAWA recommends reading the Cortex Recognition manual for details on this procedure.
- In use, the robot moves into a position (or executes a searching motion) so the camera captures the part.
- The robot job sends an I/O command to the Cortex Recognition program to trigger the camera.
- The camera sends an image back to the Cortex Recognition program.
- The program decides if it recognizes any of the taught parts and identifies which one it sees and the six degrees (X, Y, Z, Rx, Ry, and Rz) deviation the parts orientation is in relation to the taught position. This deviation displays on the monitor (if there is one) and relays back to a position variable in the robot job.
- Inside a loop in a running job, the robot will then attempt to apply the deviation as a correction to move to the trained position relative to the part. When this iterative process results in a sufficiently small deviation, the position and orientation of the part relative to the camera (and robot) is known. The robot will then be able to (typically) pick the part.

## 1.1 Reference to Other Documentation

For additional information refer to the following:

- **YRC1000 Robot Controller**
  - *READ FIRST!! Safety Requirements (179526-1CD)*
  - *YASKAWA YRC1000 Instruction Manual (RE-CTO-A221)*
  - *YRC1000 Concurrent I/O Parameter Manual (RE-CKI-A467)*
  - *YRC1000 Maintenance Manual (RE-CHO-A114)*
  - *YRC1000 Alarm Codes (RE-CER-A600)*
  - Robot Manual for your robot model
  - *YRC1000 General Operators Manual (RE-CSO-A051)*
  - *YRC1000 Arc Welding Operators (RE-CSO-A052)*
  - *YRC1000 Spot Welding using Motor Gun Operators (RE-CSO-A054)*
  - *YRC1000 Spot Welding using Air Gun Operators (RE-CSO-A055)*
  - *YRC1000 Material Handling, Press Tending, Cutting, and Other Applications, Operators (RE-CSO-A053)*
  - *YRC1000 Painting Non-Purge, Operator (RE-CSO-A057)*
  - Vendor manuals for system components not manufactured by YASKAWA.

- **YRC1000micro Robot Controller**
  - *READ FIRST!! Safety Requirements (183441-1CD)*
  - *YRC1000micro Instructions (RE-CTO-A222)*
  - *YRC1000micro Concurrent I/O Parameter Manual (RE-CKI-A469)*
  - *YRC1000micro Maintenance Manual (RE-CHO-A115)*
  - *YRC1000micro Alarm Codes (RE-CER-A601)*
  - Robot Manual for your robot model
  - *YRC1000micro General Operators Manual (RE-CSO-A058)*
  - Vendor manuals for system components not manufactured by YASKAWA.

# 2 Robot Setup

The MS3D Cortex YRC MotoPlus Application is setup with default configuration described in this section. If requiring a different configuration refer to *chapter 3 “Custom Configuration”*.

## 2.1 Robot Communication

By default, the Robot Controller will attempt to communicate with the Cortex PC with the following IP configuration:

– **Cortex IP address:** 192.168.1.30

– **Port:** 9000

## 2.2 Robot User Frame and Variables

By default, the MS3D Cortex uses the following robot controller variables:

| Variable | Name | Description |
|---|---|---|
| UF#5 | Robot User Frame to Ref | The robot user frame referenced by the P-Var that holds the location result. |
| P000 | P Variable for Results | The robot position-variable that holds the result of the vision inspection. |
| I000 | I Variable Search Model | The integer variable that supplies the model ID number when specifying a search model to the Cortex Vision System. |
| I001 | I Variable Found Model | The integer variable that holds the model ID found on the last vision inspection. |
| I002 | I Variable Error Code | The integer-variable that holds any returned error code from the last vision inspection. |

## 2.3 Robot I/O

The Robot I/O's below are default address values.

**NOTICE**

The standard robot ladder has been modified to map results to the appropriate robot inputs.

| Address | Robot Output | | Description |
|---|---|---|---|
| 10100 | 73 | Trigger RI | |
| 10101 | 74 | Trigger RL | |
| 10102 | 75 | Light ON | |
| 10103 | 76 | Light OFF | |
| 10104 | 77 | Send Integer | Sends robot integer variable value alone |
| 10105 | 78 | | |
| 10106 | 79 | | |
| 10107 | 80 | | |
| 10110 | 81 | Data Valid | Inspection Complete |
| 10111 | 82 | Object Found | |
| 10112 | 83 | Object Not Found | |
| 10113 | 84 | OOBXYZ | (Object out of bounds) |
| 10114 | 85 | OOBRxRyRz | (Object out of bounds) |
| 10115 | 86 | Out of Focus | |
| 10116 | 87 | | |
| 10117 | 88 | | |

- **Trigger RI** - When Cortex Recognition receives an RI command it recognizes the objects and sends the guidance coordinates without getting a new image from the camera.
- **Trigger RL** - When Cortex Recognition receives an RL command it will get a new image from the camera, by recognizing the objects and sending the guidance coordinates.
- **Light ON** - Turns on the integrated camera light, if applicable.
- **Light OFF** - Turns off the integrated camera light, if applicable.
- **Send Integer** - Sends the value of the robot integer variable without a trigger command.

■ **If the robot integer variable search model contains:**

- A value other than 0, the RI and RL commands will pass that value to the Cortex Recognition system specifying that it search for that model.
- A value of 0, is a standard search that takes place looking for any taught model.

## 2.4 Example of Triggering Vision System

The “RI” command triggers an inspection in the vision system. Using the robot variables in the configuration file, can trigger a search for a specific image model. Additionally, the trigger can search all image models and return the closest match.

The value in the robot integer variable for “search model,” controls the trigger method.”

| NOTICE |
| --- |
| See the Cortex Recognition user manual for more information on commands to the vision system. |

- **Example:**
  I000 is designated as the search model variable

  I000 = 12

  Output 74 is turned on

  The actual command sent is “rI0012”. This will search specifically for image model #12.

  I000 = 0

  Output 74 is turned on

  The actual command sent is “RI”. This searches through all image models and returns the closest match.

# 3 Custom Configuration

If the MS3D Cortex standard setup needs customization, a configuration file can be created and stored on a USB drive connected to the Robot Controller.

## 3.1 Create Configuration File

Create a configuration file that is to be loaded onto USB drive connected to the controller.

The file can be created/edited directly in a standard text editor and saved as an ASCII file (plain text) with the name “Cortex.cfg”. The format of the ASCII file is as follows:

| | |
|---|---|
| 192.168.255.5\CrLf | IP Address |
| 9000\CrLf | Port |
| 5\CrLf | User Frame |
| 0\CrLf | Robot position variable index for output |
| 0\CrLf | Robot integer variable (I) index for search model |
| 1\CrLf | Robot interger variable (I) index of model found |
| 2\CrLf | Robot interger variable (I) index for error code |
| 10100\CrLf | Starting address for I/O commands |
| 10110\CrLf | Starting address for I/O results |

### 3.1.1 Parameters That Can Be Set Or Modified

- **Cortex IP Address** - This is the IP address of the embedded PC port that is connected to the robot.
- **Port** - Defined by Cortex Vision System. Default value is 9000.
- **Robot User Frame to Ref** - The robot user frame referenced by the PVar that holds the location result.
- **P Variable for Results** - The index of the robot position-variable that will hold the result of the vision inspection.
- **I Variable Search Model** - The index of robot integer-variable that will supply the model ID number when specifying a search model to the Cortex Vision System.
- **I Variable Found Model** - The index of the robot integer-variable that will hold the model ID found on the last vision inspection.
- **I Variable Error Code** - The index of the robot integer-variable that will hold any returned error code from the last vision inspection.
- **Commands Starting Address** - This is the starting address for the robot I/O commands.
- **Results Starting Address** - The inspection status signals are returned in robot outputs. This is the starting address for those results.

## 3.2 Saving the Configuration File to Robot Controller

Save the “Cortex.cfg” to the root of a USB drive (not included.) Attach the USB drive to the Robot Controller

– **YRC1000** - Connect the USB drive inside the Robot Controller on the ACP01 board at the CN102 port.

– **YRC1000micro** - Connect the USB drive to CN102 on the front of the Robot Controller.

**NOTICE**

- Permanently leave the USB drive connected to the Robot Controller.

The Robot Controller reads the configuration file at each start-up.

Cycle Power on the Robot Controller for the new settings to take effect.

# 4 Troubleshooting

If the delay time is not long enough for the camera(s) to make a connection before launching the Cortex application try one of these fixes;

- **Disable any unused TCP/IP ports.**
  (1) Right click “My Network Neighborhood”.
  (2) Right click the unused port.
  (3) Select “Disable”.
  (4) Repeat for each unused port.

- **Adjust the Cortex application start delay time.**
  (1) Launch the Delayed Start application and press 'Stop'.
  (2) Increase the time value by clicking on the seconds count using the up/down controls to set the new delay time.
  (3) Click 'Save'.
  (4) Close the application.
  (5) Reboot to test the new time value.

- **Vendor Resources:**
  - UsersGuide_Baumer_GigE_IPConfig_Win_v30e_100913.pdf (located in \Program Files\IpConfig)
  - CortexRecognition.pdf (accessible from Programs -> Recognition Robotics, Inc. selection)

# YRC1000 & YRC1000micro OPTIONS
# INSTRUCTIONS

## FOR: MS3D CORTEX CONFIGURATION

**For inquiries or after-sales service on this product, contact your local YASKAWA representative as shown below.**


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Specifications are subject to change without notice for ongoing product modifications and improvements.

YASKAWA


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